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LAB:ROLE-BASED ACCESS CONTROL

1.1 INTRODUCTION

I started the lab with a required lab set up.I clicked on ctrl alt and delete to bring up the login page,signed in as admin and then downloaded the lab files.

1.2 LAB 1:ROLE -BASED ACCESS CONTROL

LAB SCENARIO

I was asked to create a proof of concept showing how Azure users and groups are created. Also, how role-based access control is used to assign roles to groups.

EXERCISE 1:CREATE THE SENIOR ADMINS GROUP WITH THE USER ACCOUNT JOSEPH PRICE AS ITS MEMBER.

In this task,I used the Azure portal to create a user account for Joseph Price and a senior Admins group and added the user account of Joseph Price to the group.

TASK 1

In this task;I created a user account for Joseph Price.

Step 1

I started a browser session and signed-in to the Azure portal using the credentials provided,

Step 2

I searched for Microsoft Entra ID in the search resources, services, and docs text box at the top of the Azure portal page.

Step 3

On the Overview blade of the Microsoft Entra ID tenant, in the manage section, I selected users, and new user.

Step 4,5,6,7 and 8

On the New User blade, I ensured that the Create user option was selected, and specified the provided settings.The new user was already created, I refreshed the users blade to verify the user was created and then proceeded to task 2.

TASK 2

In this task, I created the Senior Admins group, added the user account of Joseph Price to the group, and configured it as the group owner.

Step 1

In the Azure portal,I navigated back to the blade displaying Microsoft Entra ID tenant.

Step 2

In the Manage section, I clicked on Groups, and then selected + New group.

Step 3

On the New Group blade, I specified the settings provided and left the others with their default values.

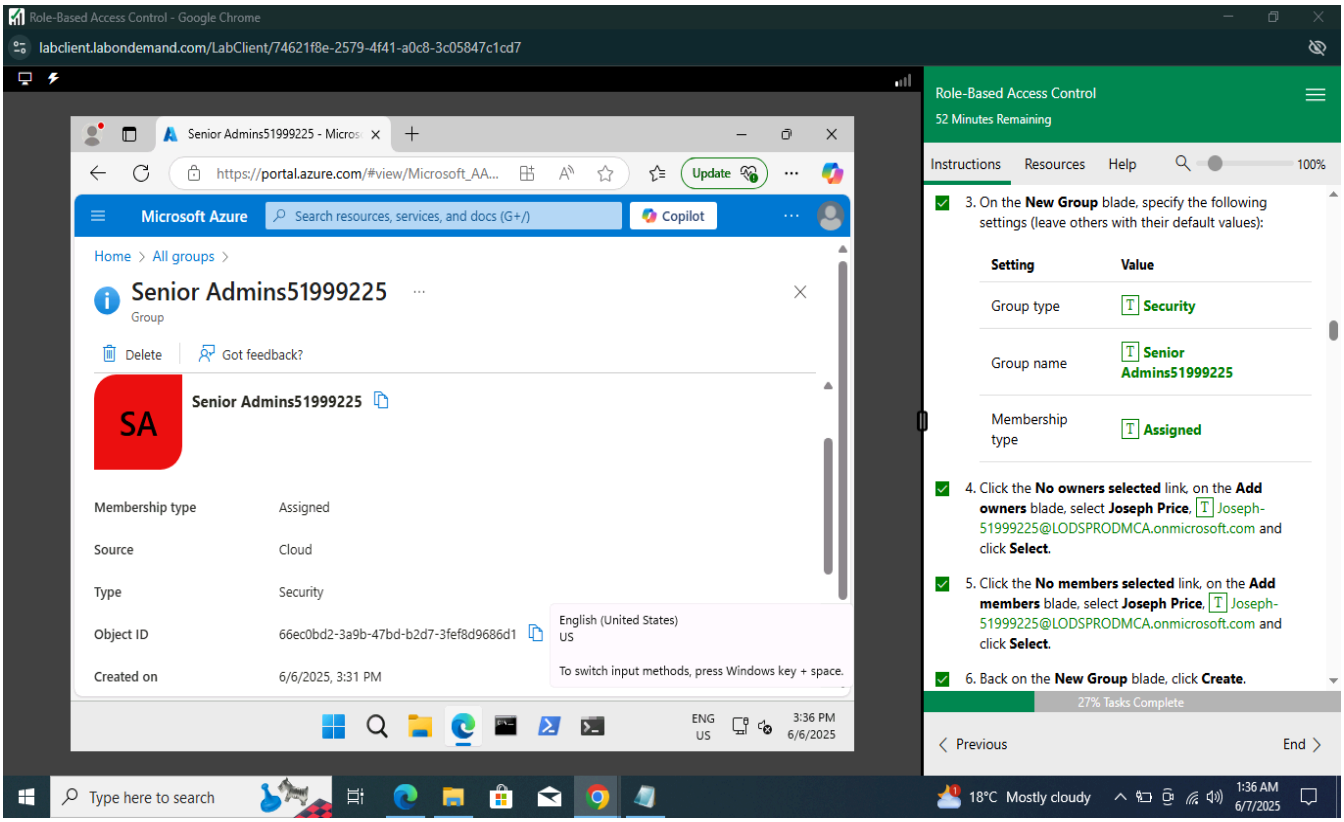
Step 4

I clicked on the No owners selected link and No members selected link and selected Joseph Price.

Step 5

I then clicked on create to finish creating group

Here is my screenshot



EXERCISE 2:CREATE A JUNIOR ADMINS GROUP CONTAINING THE USER ACCOUNT OF ISABEL GARCIA AS ITS MEMBER

In this exercise,I used PowerShell to create a user account for Isabel Garcia, the Junior Admins group and added the user account of Isabel Garcia to the group.

TASK 1

In this task, I created a user account for Isabel Garcia by using PowerShell.

Step 1

I opened the Cloud Shell by clicking the first icon in the top right of the Azure Portal.When prompted, I selected PowerShell and created storage.

Step 2

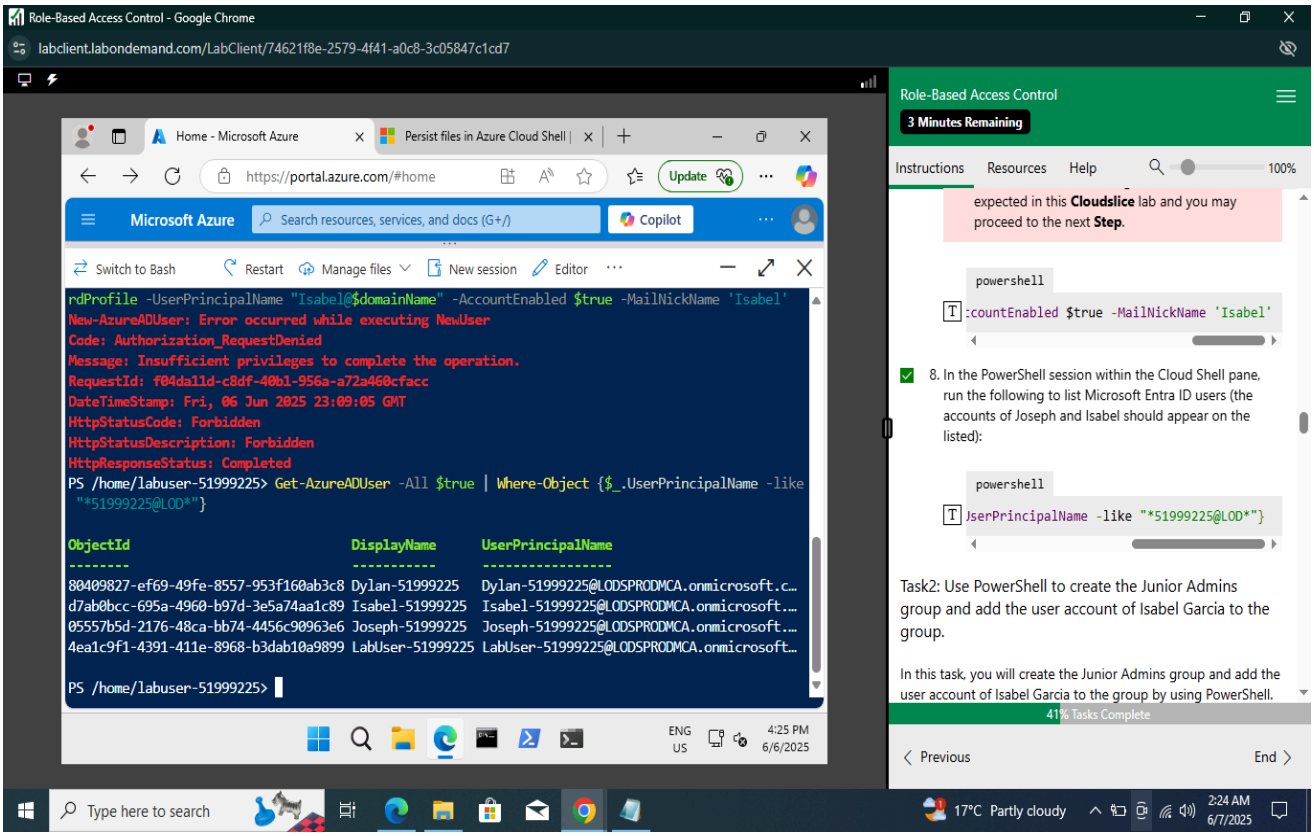
I ensured PowerShell was selected in the drop-down menu in the upper-left corner of the Cloud Shell pane.

Step 3,4,5,6,7

In the powershell session within the Cloud Shell pane,I ran the command; **$passwordProfile = New-Object -TypeName Microsoft.Open.AzureAD.Model.PasswordProfile** to create a password profile object, **$passwordProfile.Password = "Pa55w.rd1234"**to set the value of the password within the profile object, **Connect-AzureAD** to connect to Microsoft Entra ID, **$domainName = ((Get-AzureAdTenantDetail).VerifiedDomains)[0].Name** to identify the name of your Microsoft Entra tenant and **New-AzureADUser -DisplayName 'Isabel Garcia' -PasswordProfile $passwordProfile -UserPrincipalName "Isabel@$domainName" -AccountEnabled $true -MailNickName 'Isabel'** to create a user account for Isabel Garcia.

Step 8

I then ran the command,**Get-AzureADUser -All $true | Where-Object {$_.UserPrincipalName -like "*52006700@LOD*"}** to list Microsoft Entra ID users and the accounts of Joseph and Isabel appeared on the list.Here is a screenshot



TASK 2

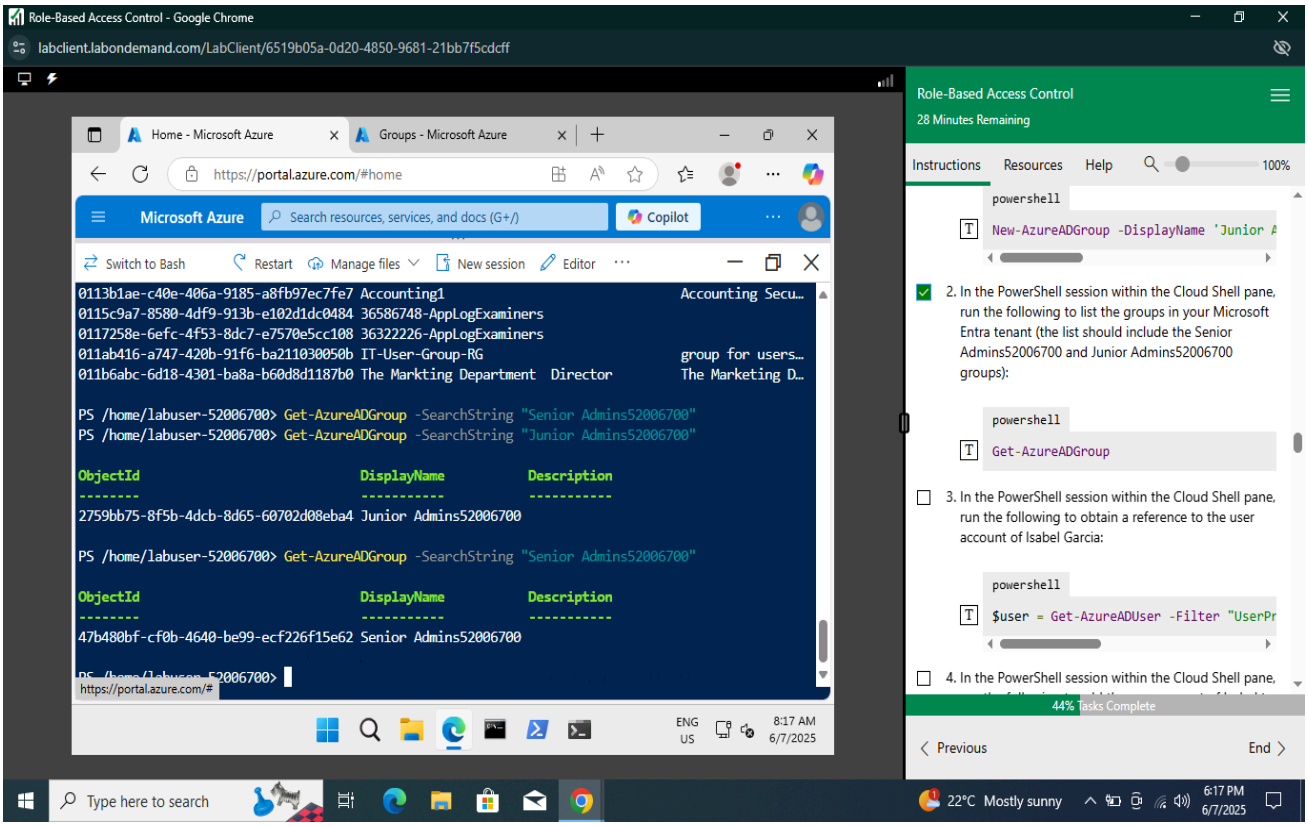
In this task, I created the Junior Admins group and added the user account of Isabel Garcia to the group by using PowerShell.

Step 1

In the same PowerShell session from the previous task within the Cloud Shell pane, I ran the command “New-AzureADGroup -DisplayName 'Junior Admins52006700' -MailEnabled $false -SecurityEnabled $true -MailNickName JuniorAdmins” to create a new security group named Junior Admins

Step 2

I ran the command “Get-AzureADGroup” to list the groups in Microsoft Entra tenant.The list included the Senior Admins52006700 and Junior Admins52006700 groups.Here is my screenshot



Step 3

I ran the command, $user = Get-AzureADUser -Filter "UserPrincipalName eq 'Isabel-52006700@LODSPRODMCA.onmicrosoft.com'" to obtain a reference to the user account of Isabel Garcia:

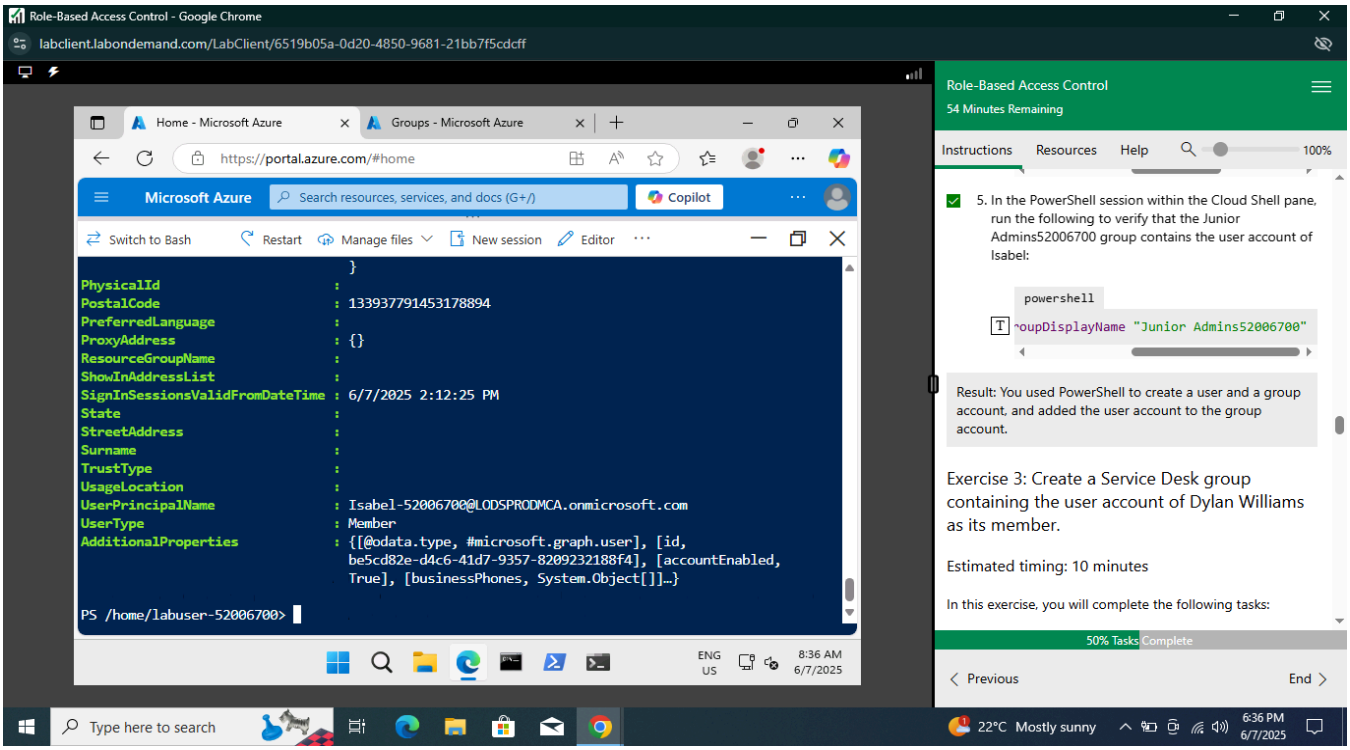
Step 4

I ran the command “Add-AzADGroupMember -MemberUserPrincipalName $user.userPrincipalName -TargetGroupDisplayName "Junior Admins52006700" “ to add the user account of Isabel to the Junior Admins52006700 group.

Step 5

In the PowerShell session within the Cloud Shell pane, I ran the command, Get-AzADGroupMember -GroupDisplayName "Junior Admins52006700" to verify that the Junior Admins52006700 group contains the user account of Isabel:

Here is my screenshot



EXERCISE 3:CREATE A SERVICE DESK GROUP CONTAINING THE USER ACCOUNT OF DYLAN WILLIAMS AS ITS MEMBER

In this exercise, I used Azure CLI to create a user account for Dylan Williams, the Service Desk group and added the user account of Dylan to the group.

TASK 1

In this task, I created a user account for Dylan Williams.

Step 1

In the drop-down menu in the upper-left corner of the Cloud Shell pane, I selected Bash, and clicked Confirm.

Step 2

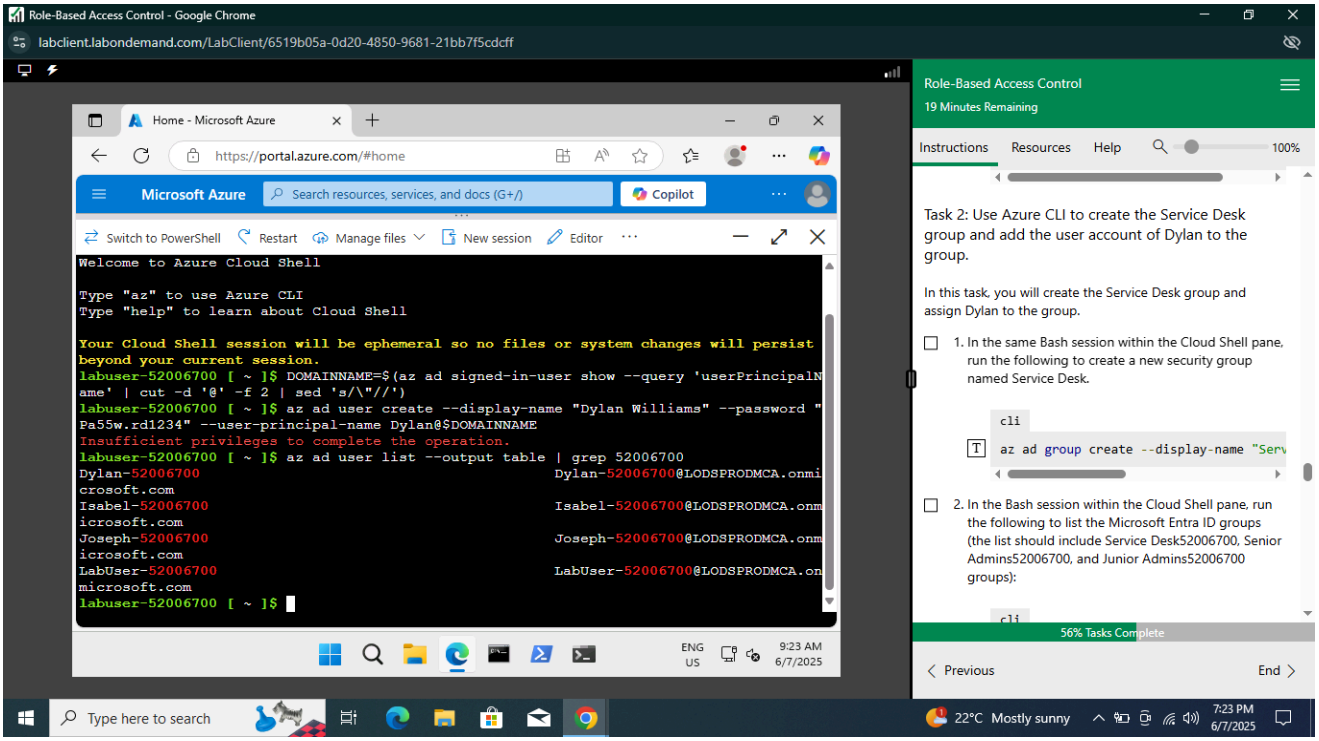
In the Bash session within the Cloud Shell pane,I ran the command,DOMAINNAME=$(az ad signed-in-user show --query 'userPrincipalName' | cut -d '@' -f 2 | sed 's/\"//') to identify the name of Microsoft Entra tenant.

Step 3

In the Bash session, I ran the command, az ad user create --display-name "Dylan Williams" --password "Pa55w.rd1234" --user-principal-name Dylan@$DOMAINNAME to create a user, Dylan Williams.

Step 4

In the Bash session,I ran the command, az ad user list --output table | grep 52006700 to list Microsoft Entra ID user accounts, the list included user accounts of Joseph, Isabel, and Dylan.Here is my screenshot



TASK 2

In this task,I created the Service Desk group and assigned Dylan to the group

Step 1

In the Bash session within the Cloud Shell pane,I ran the command, az ad group create --display-name "Service Desk52018238" --mail-nickname "ServiceDesk" to create a new security group named Service Desk.

Step 2

In the Bash session, I ran the command, az ad group list -o table to list the Microsoft Entra ID groups. The list included Service Desk52018238, Senior Admins52018238, and Junior Admins52018238 groups.

Step 3

In the Bash session,I ran the following;USER=$(az ad user list --filter "UserPrincipalName eq 'Dylan-52018238@LODSPRODMCA.onmicrosoft.com'") to obtain a reference to the user account of Dylan Williams.

Step 4

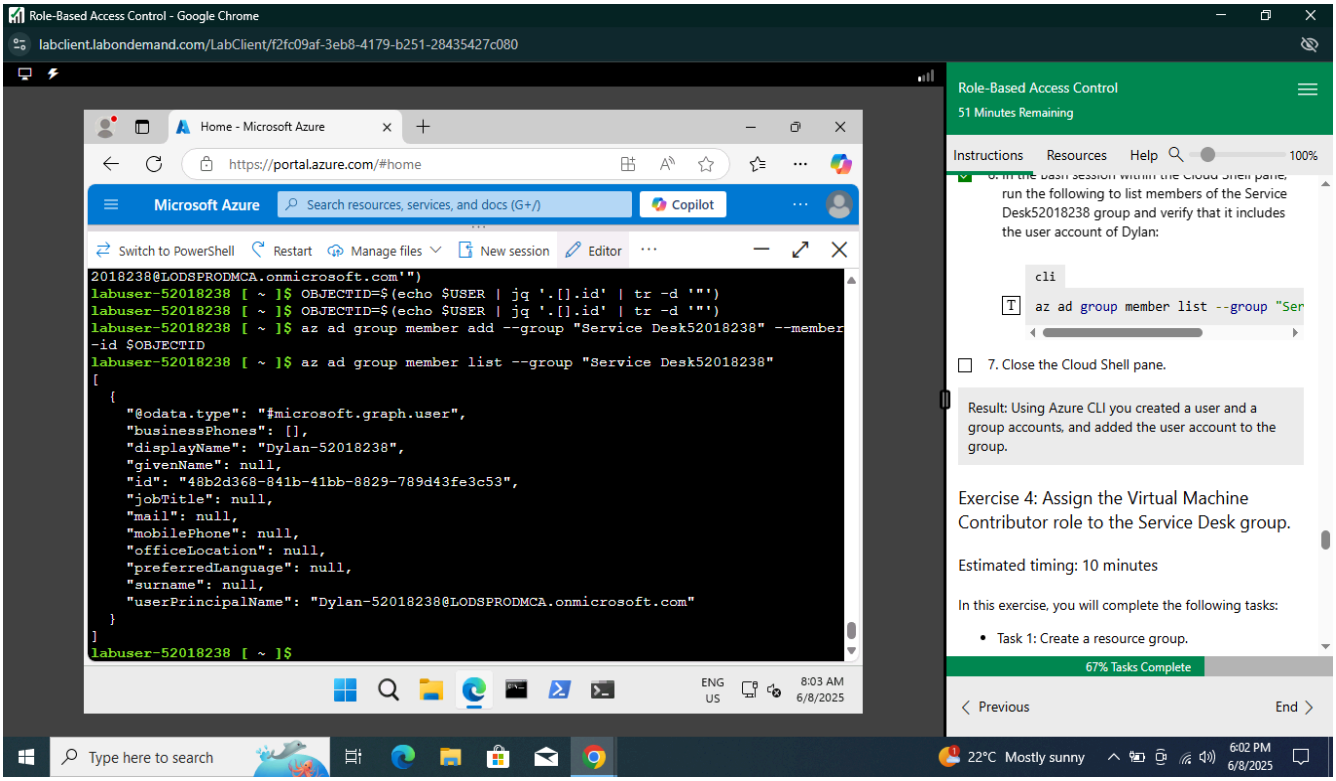
In the Bash session, I ran the command; OBJECTID=$(echo $USER | jq '.[].id' | tr -d "") to obtain the objectId property of the user account of Dylan Williams

Step 5

In the Bash session, I ran the following command;az ad group member add --group "Service Desk52018238" --member-id $OBJECTID to add the user account of Dylan to the Service Desk52018238 group:

Step 6

In the Bash session,I ran the command;az ad group member list --group "Service Desk52018238" to list members of the Service Desk52018238 group and verified that it included the user account of Dylan.Here is my screenshot



Step 7

I closed the cloud shell and moved to the next task.

EXERCISE 4:ASSIGN THE VIRTUAL MACHINE CONTRIBUTOR ROLE TO THE SERVICE DESK GROUP

In this exercise,I created a resource group and assigned the Service Desk Virtual Machine Contributor permissions to the resource group.

TASK 1

In this task,I created a resource group.

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Resource groups and pressed the Enter key.

Step 2

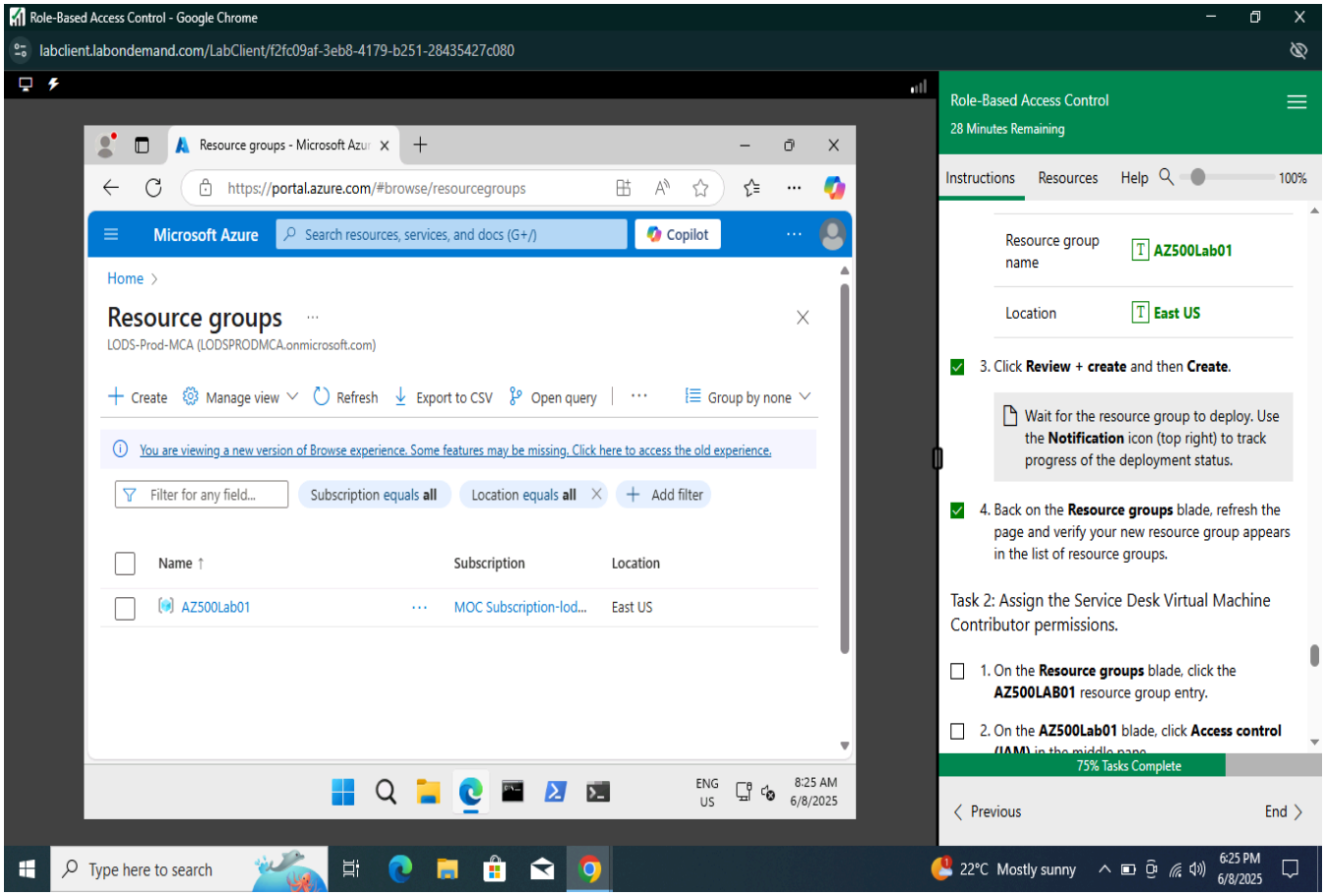
On the Resource groups blade,I clicked on + Create and specified the settings provided

Step 3

I clicked Review + create and then Create to finish creating the resource group.

Step 4

Back on the Resource groups blade,I refreshed the page and verified the new resource group appears in the list of resource groups.Here is a screenshot



TASK 2

In this task,I assigned the Service Desk Virtual Machine Contributor permissions.

Step 1

On the Resource groups blade, I clicked on the AZ500LAB01 resource group entry.

Step 2

On the AZ500Lab01 blade, I clicked on Access control (IAM) in the middle pane.

Step 3

On the AZ500Lab01 | Access control (IAM) blade,I clicked the + Add and then, in the drop-down menu, clicked Add role assignment.

Step 4

On the Add role assignment blade, I specified the settings provided and clicked Next after each step.

Step 5

I clicked on Review + assign twice to create the role assignment.

Step 6

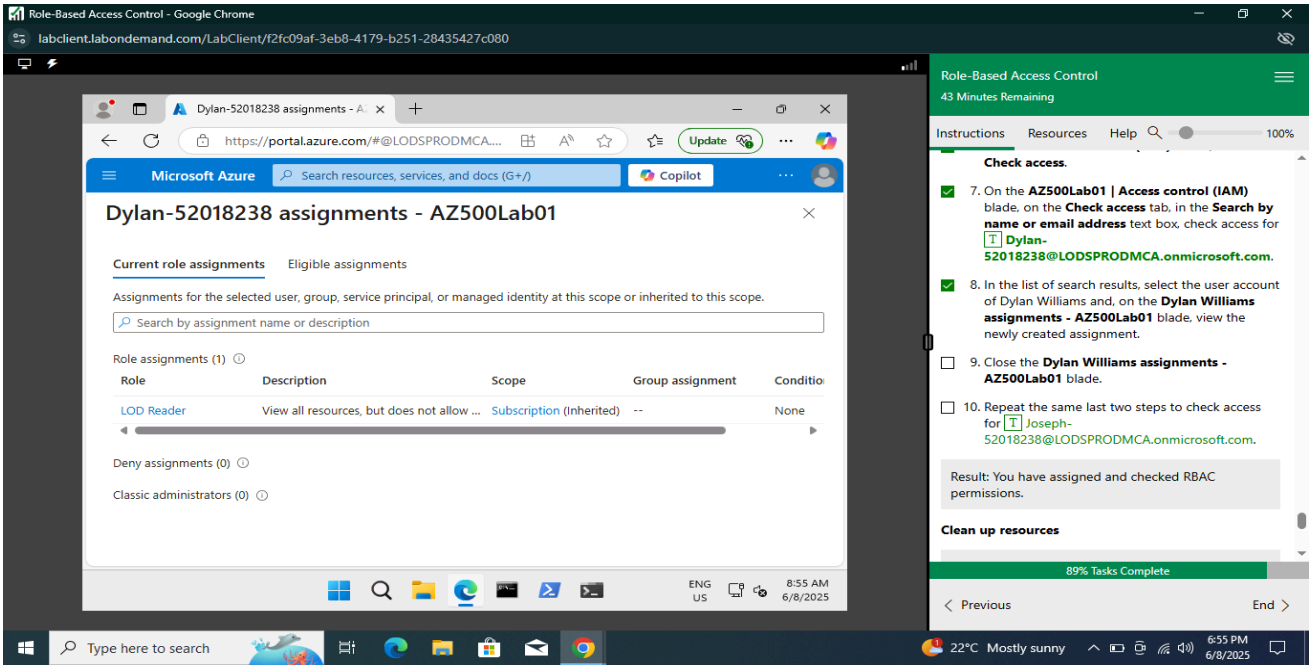
From the Access control (IAM) blade, I selected Check access.

Step 7

On the Check access tab, in the Search by name or email address text box, I checked access for Dylan.

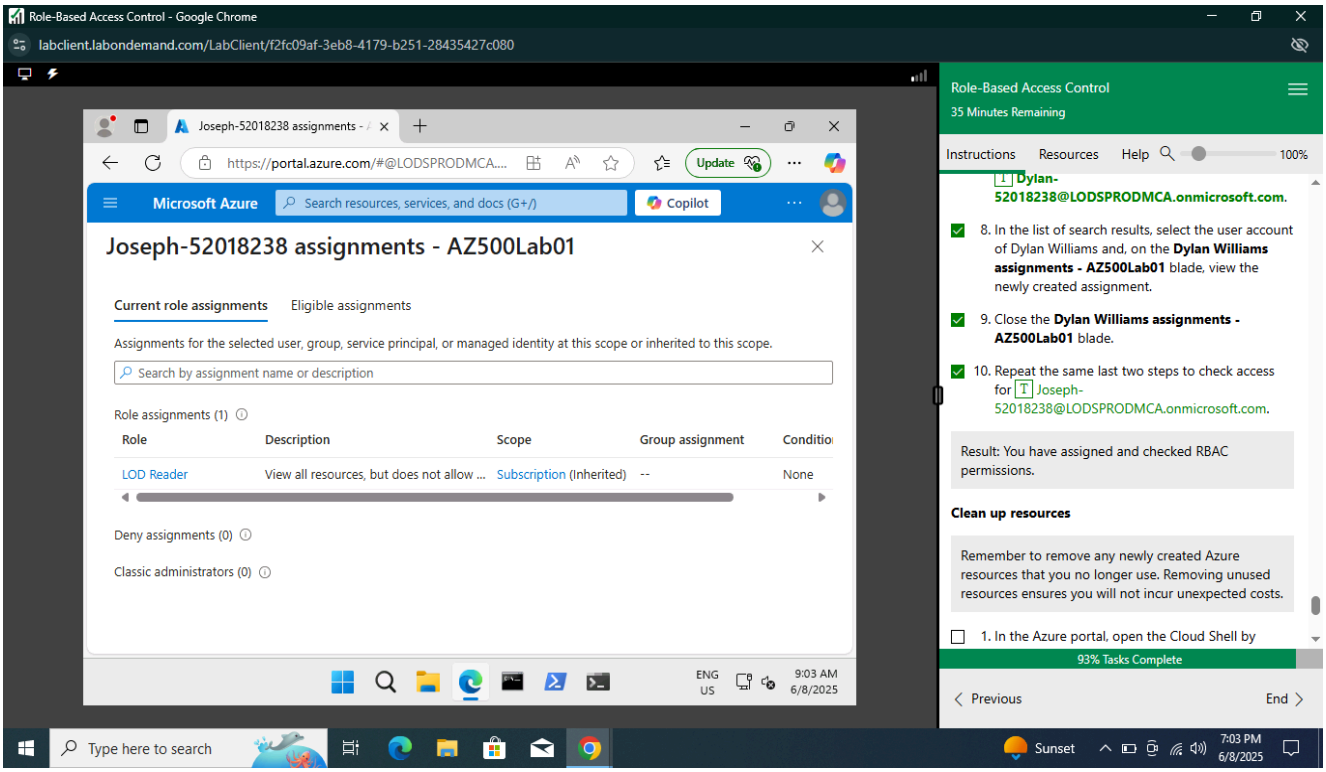
Step 8

In the list of search results, I selected the user account of Dylan Williams and, on the Dylan Williams assignments - AZ500Lab01 blade, viewed the newly created assignment.Here is my screenshot



Step 9 and 10

I closed the Dylan Williams assignments - AZ500Lab01 blade and repeated the same last two steps to check access for joseph.Here is my screenshot



CLEAN UP RESOURCES

I cleaned up my resources and completed the lab.